

Beyond the Large N Limit: Non-linear $W(\infty)$ as symmetry of the $SL(2,R)/U(1)$ coset model

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Abstract

We show that the symmetry algebra of the $SL(2,R)_{\{k\}}/U(1)$ coset model is a non-linear deformation of $W_{\{\infty\}}$, characterized by k . This is a universal W -algebra which linearizes in the large k limit and truncates to $W_{\{N\}}$ for $k=-N$. Using the theory of non-compact parafermions we construct a free field realization of the non-linear $W_{\{\infty\}}$ in terms of two bosons with background charge. The W -characters of all unitary $SL(2,R)/U(1)$ representations are computed. Applications to the physics of 2-d black hole backgrounds are also discussed and connections with the KP approach to $c=1$ string theory are outlined.

1 References And Citations

This sample cites source keys [1] and [2]. Section 1 and Equation 1 provide cross-reference coverage.

$$a^2 + b^2 = c^2 \tag{1}$$

References

- [1] Source bibliography entry 1 extracted from arXiv metadata/source key '10'.
- [2] Source bibliography entry 2 extracted from arXiv metadata/source key '11'.
- [3] Source bibliography entry 3 extracted from arXiv metadata/source key '12'.
- [4] Source bibliography entry 4 extracted from arXiv metadata/source key '13'.